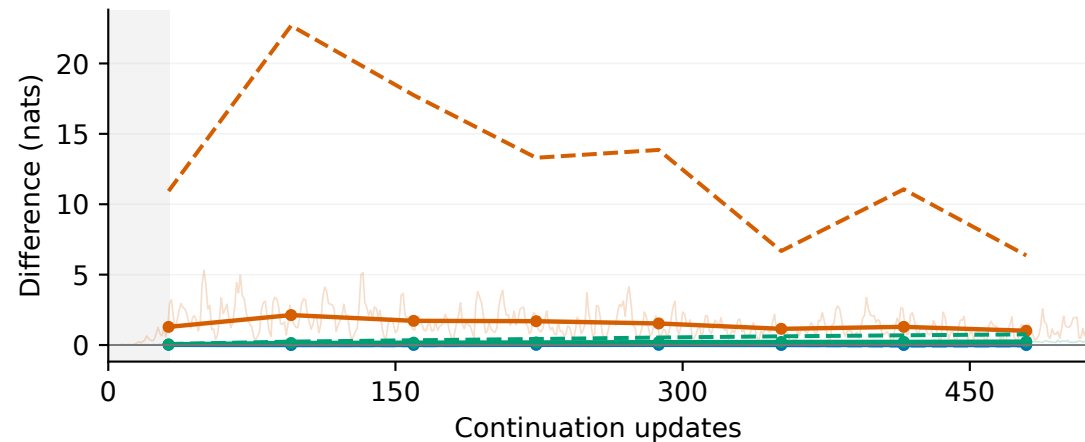
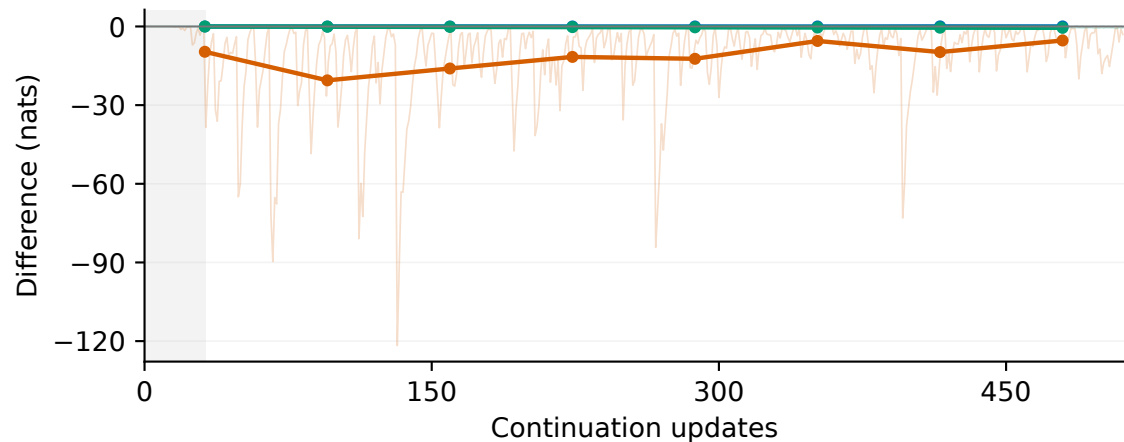


—●— SGD – Heavy Ball effective $h=0.0003$
—●— SGD – Heavy Ball effective $h=0.0012$
—●— Adam – Head only (different settings) raw $\eta=1.27\text{e-}05 / 0.0003$

(a) Paired measured response and reconstruction



(b) Paired remaining response $\Delta\epsilon$



Light lines: observed values. Dots: 64-step means. Panel (a): solid ΔL ; dashed $\Delta A + \Delta Q_2$.
 Common difficulty cancels. Original shared step axis; no fitted time alignment.
 Initial registered protocol.